

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 2)

If $y = \frac{(\sqrt{x}+1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}(3\cos^2 x - 5)\cos^3 x$, then $96y' \left(\frac{\pi}{6} \right)$ is equal to :

Q2 - 2024 (27 Jan Shift 1)

Let for a differentiable function $f : (0, \infty) \rightarrow R$, $f(x) - f(y) \geq \log_e \left(\frac{x}{y} \right) + x - y, \forall x, y \in (0, \infty)$. Then

$\sum_{n=1}^{20} f' \left(\frac{1}{n^2} \right)$ is equal to _____

Q3 - 2024 (27 Jan Shift 1)

Let $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3), x \in R$.

Then $f'(10)$ is equal to _____

Q4 - 2024 (29 Jan Shift 1)

Suppose

$$f(x) = \frac{(2^x + 2^{-x}) \tan x \sqrt{\tan^{-1}(x^2 - x + 1)}}{(7x^2 + 3x + 1)^3}$$

Then the value of $f'(0)$ is equal to

- (1) π
- (2) 0
- (3) $\sqrt{\pi}$
- (4) $\frac{\pi}{2}$

Q5 - 2024 (29 Jan Shift 2)

Let $y = \log_e \left(\frac{1-x^2}{1+x^2} \right), -1 < x < 1$. Then at $x = \frac{1}{2}$, the value of $225(y' - y'')$ is equal to

- (1) 732
- (2) 746
- (3) 742
- (4) 736

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Answer Key

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Solutions

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Q1

$$y = \frac{(\sqrt{x} + 1)(x^2 - \sqrt{x})}{x\sqrt{x} + x + \sqrt{x}} + \frac{1}{15}(3\cos^2 x - 5)\cos^3 x$$

$$y = \frac{(\sqrt{x} + 1)(\sqrt{x})((\sqrt{x})^3 - 1)}{(\sqrt{x})((\sqrt{x})^2 + (\sqrt{x}) + 1)} + \frac{1}{5}\cos^5 x - \frac{1}{3}\cos^3 x$$

$$y = (\sqrt{x} + 1)(\sqrt{x} - 1) + \frac{1}{5}\cos^5 x - \frac{1}{3}\cos^3 x$$

$$y' = 1 - \cos^4 x \cdot (\sin x) + \cos^2 x (\sin x)$$

$$y' \left(\frac{\pi}{6} \right) = 1 - \frac{9}{16} \times \frac{1}{2} + \frac{3}{4} \times \frac{1}{2}$$

$$= \frac{32 - 9 + 12}{32} = \frac{35}{32}$$

$$= 96y' \left(\frac{\pi}{6} \right) = 105$$

Q2

$$f(x) - f(y) \geq \ln x - \ln y + x - y$$

$$\frac{f(x) - f(y)}{x - y} \geq \frac{\ln x - \ln y}{x - y} + 1$$

Let $x > y$

$$\lim_{y \rightarrow x} f'(x^-) \geq \frac{1}{x} + 1 \dots (1)$$

Let $x < y$

$$\lim_{y \rightarrow x} f'(x^+) \leq \frac{1}{x} + 1 \dots (2)$$

$$f^1(x^-) = f^1(x^+)$$

$$f^1(x) = \frac{1}{x} + 1$$

$$f' \left(\frac{1}{x^2} \right) = x^2 + 1$$

$$\sum_{x=1}^{20} (x^2 + 1) = \sum_{x=1}^{20} x^2 + 20$$

$$= \frac{20 \times 21 \times 41}{6} + 20$$

$$= 2890$$

Q3

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Solutions

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$$f(x) = x^3 + x^2 \cdot f'(1) + x \cdot f''(2) + f'''(3)$$

$$f'(x) = 3x^2 + 2x f'(1) + f''(2)$$

$$f''(x) = 6x + 2f'(1)$$

$$f'''(x) = 6$$

$$f'(1) = -5, f''(2) = 2, f'''(3) = 6$$

$$f(x) = x^3 + x^2 \cdot (-5) + x \cdot (2) + 6$$

$$f'(x) = 3x^2 - 10x + 2$$

$$f'(10) = 300 - 100 + 2 = 202$$

Q4

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(2^h + 2^{-h}) \tan h \sqrt{\tan^{-1}(h^2 - h + 1)} - 0}{(7h^2 + 3h + 1)^3 h}$$

$$= \sqrt{\pi}$$

Q5

$$y = \log_e \left(\frac{1-x^2}{1+x^2} \right)$$

$$\frac{dy}{dx} = y' = \frac{-4x}{1-x^4}$$

Again,

$$\frac{d^2y}{dx^2} = y'' = \frac{-4(1+3x^4)}{(1-x^4)^2}$$

Again

$$y' - y'' = \frac{-4x}{1-x^4} + \frac{4(1+3x^4)}{(1-x^4)^2}$$

$$\text{at } x = \frac{1}{2},$$

$$y' - y'' = \frac{736}{225}$$

$$\text{Thus } 225(y' - y'') = 225 \times \frac{736}{225} = 736$$

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